EPFL - Computer Science

COM-402 Information Security and Privacy

Lecture 12 — December 8, 2025

Author: Jonathan Dutoit

Professor: Mathias Payer

ML Security

Learning goals

- Basics of machine learning security
- Stealing models and privacy implications
- How the output of models can be altered
- Pitfalls with machine learning setups

Machine Learning (ML)

Definition (Wikipedia)

Machine learning [...] gives "computers the ability to learn without being explicitly programmed" [and] [...] explores the study and construction of algorithms that can learn from and make predictions on data – such algorithms overcome following strictly static program instructions by making data-driven predictions or decisions, through building a model from sample inputs.

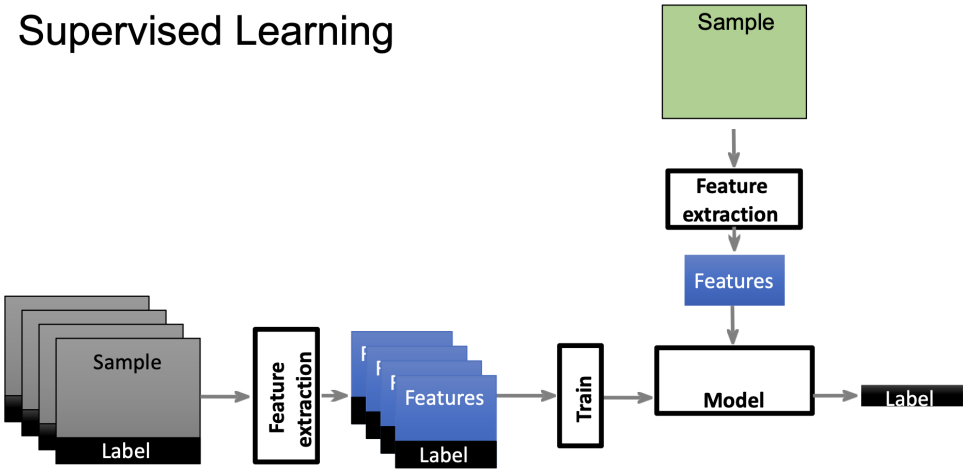
Machine Learning is becoming Ubiquitous, it's everywhere! Machine learning can be separated into 3 main categories:

- Unsupervised learning
- **Supervised learning**
- Reinforcement learning

And

- Labelled data
- Direct feedback
- Predict outcome/future
- Example: estimate a house price

LLMs are exploding and security research for LLMs is quickly gaining attention.



Adversary Goals

- Steal information: **privacy issue**
- Steal the model: **intellectual property**
- Alter the output: **fooling the user**

Model Stealing

Protecting intellectual property in ML

Machine Learning Under Adversarial Conditions

Up there are the more threatening models, and down is the higher success rates Opaque-box attacks

- Model architecture and parameters unknown
- Can only interact *blindly* with the model

Grey-box attacks

- Model architecture known, parameters unknown
- Can only interact with the model, but has information about they type of the model

Clear-box attacks

- Know architecture and parameters
- Can replicate the model and use the model's internal parameters in the attack

Machine Learning as a Service

1. The Cloud, e.g., Amazon ML or Google Prediction API, (pre-)trains a classifier using their own data.

2. Make this classifier available as a service for users to query

3. The user makes a query: "Given their profile (photos, posts, metadata), what pet has this Facebook user?

Model Stealing

Confidentiality of the model itself (e.g., intellectual property)

Good ML models require considerable investment:

- Collecting data takes time and money
- Training infrastructure is expensive (time and $$$)

Goal: "steal" the expensive model by observing its outputs with lower cost than obtaining the data and training.

Stealing a linear model

Assume adversary knows the model is a linear architecture (*Grey-box model*).

Assume x is two-dimensional:

$$f(x) = w_1 x_1 + w_2 x_2 + b.$$

Adversary's goal: steal parameters w, b

How many input-output pairs (x, f(x)) the adversary must observe to steal the model?

For two-dimensional x only need 3 queries!

In general, if a linear model users *d features*, the adversary needs **d+1** different queries to steal by solving the linear system for w, b

Stealing a non-linear model

Assume adversary know the model's architecture (*Grey-box model*)

$$f_w(x).$$

Adversary's goal: steal parameters w

Retraining attack

Observe many queries X = *(x, f(x))*, and fit the model on X like on any other training data!

Takes many queries. For a neural network with 2K parameters, need 11K queries to get 99.9% similarity.

More recent work has reduced these numbers.

More on stealing neural networks

"Given (oracle) query access to a neural network, learned through stochastic gradient descent, can we extract a functionality equivalent model?"

Implication of this attack against neural networks

Achievement: extract a 100,000 parameter neural network on the MNIST digit recognition task with $2^{21.5}$ queries in under an hour ⇒ implication from cryptographic research for ML

Assumption of the field of secure inference: observing the output of a neural network does not reveal the weights

⇒ This assumption is **false**, and therefore the field of secure inference will need to develop new techniques to protect the secrecy of trained models

Preventing model stealing

First attack = 2016, First defenses ∼ 2017

Output perturbation

Add **noise** to the probabilities output by the model to hinder reconstruction, but not accuracy

Detect suspicious queries

Identify deviations from expected on distribution of successive queries from a client

Very recent techniques! We don't yet know how robust they are

Takeaways on model stealing

- Many models are susceptible to stealing
- Complex topic, lots of ongoing research
- Crucial issue for ML as a service
- Very young field, the situation will (probably) improve!
- **Many problems**, **some solutions**

Data Stealing

Privacy Challenges in ML

Information leaks about data user for training

Testing data (or the result) might be sensitive in ML as a Service

Questions:

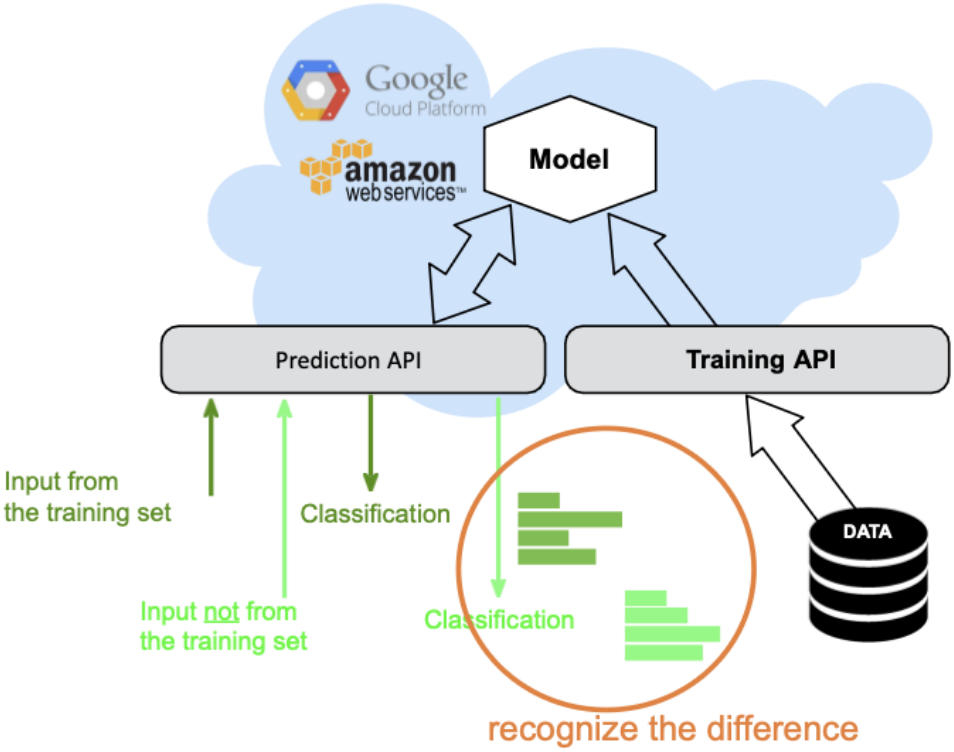
- Do trained models leak sensitive data?
- Can we train good privacy-preserving models?

Membership Inference

The attacker tries to understand what data has been used as part of the training. Is a sensitive data point more similar to the "reference" dataset or the "summary statistics"?

Membership Inference Assumptions

No knowledge of the model **parameters** (grey-box attack)



Attack

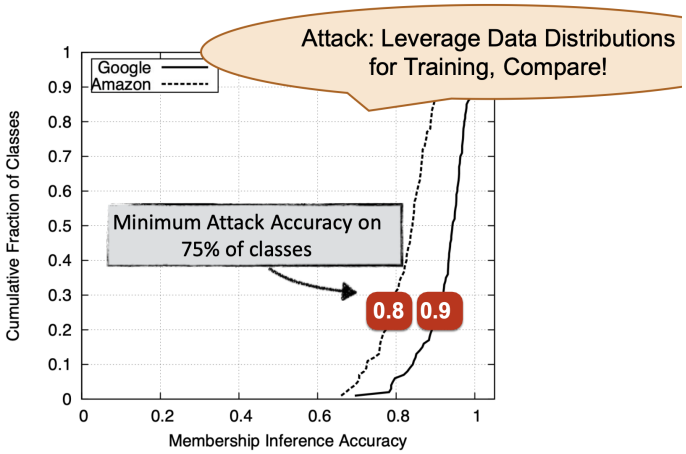
Leverage Public Data for Training, Compare! Without knowing the parameters of the actual model!

Assumptions 2.0: No samples of the training data no knowledge of the model **parameters**

Shadow Models

Try to create 'shadow' of the actual model to understand how it behave.

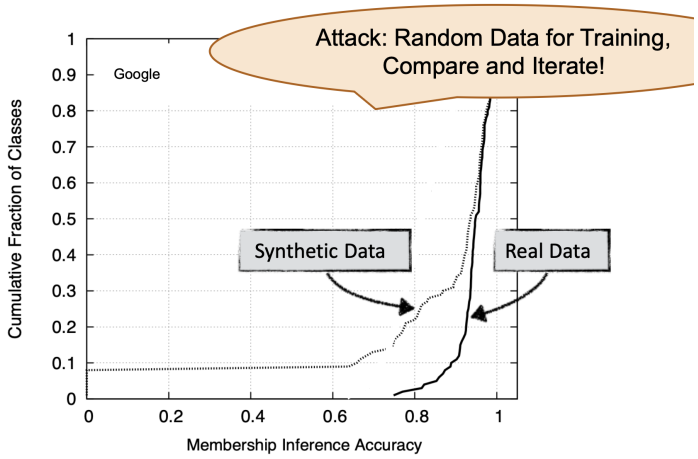
Train the attack model to predict if an input was a member of the training set (in) or a non-member (out)



Assumptions 3.0: No knowledge of the underlying distribution of the **training data** no knowledge of the model **parameters**

Data Synthesis for Shadow Models

We take random data, use the prediction API to get results and use those to assess distributions. We get very close to the precision of using real data.



Why do these attacks work?

Because the model is over-fitted on the training data!

Overfitting is the common enemy

Privacy: Does the model leak information about data in the training set?

Learning: Does the model generalize to data outside the training set?

For once, privacy and utility are not in conflict: overfitting is the common enemy

- Overfitted models leak training data
- Overfitted models lack predictive power

Need generalizability and accuracy

Takeaways on ML Privacy issues

Many problems, **some solutions**

Very young field, the situation will improve!

More research needed.

Altering the Output

Inputs that will make ML fail

Inputs to a model that an attacker has designed to cause the model to make a mistake.

Example: a classification model for animals can take a panda photo and adding noise to it (adversarial perturbation), now the models output "Gibbon" instead of "Panda". This happens because of **overfitting**.

Independent and Identically Distributed (IID) assumptions no longer hold

- *Identical*: inputs are intentionally manipulated to not belong to the training distribution.
- *Independence*: inputs are no longer draw independently; the attacker may sample from a single input repeatedly.

Machine Learning 101: Quick refresher

Objective: find model parameters that *minimize* empirical loss. How do we "train" those weights? Often, the answer is to use a (simplified) flavor of gradient descent.

Adversarial example problem

Goal: find a perturbation (adversarial noise) that *maximizes* loss

How to define similarity?

The similarity relation is often represented as *adversarial cost constraint*.

If the goal is to be imperceptible, the common cost is a norm or a perturbation.

Other similarity relations

Constraint on perturbation norm assumes that similar images are close in, e.g., Euclidean distance. This is not always true!

For example, similarity could be defined as small affine transformations (rotations, scaling, ...)

Transferability Property

"Adversarial examples have a **transferability** property:

Samples crafted to mislead a model A are likely to mislead a model B."

Attack Surface

In the most extreme case, it is possible to construct a single perturbation that will fool a model when added to any image.

Attackers need minimal resources to attack the system!

Defending against adversarial examples?

Defending n general is very head. Can only defend against a particular threat model (e.g., perturbations up to epsilon norm), and normally no guarantees.

Standard ways is *adversarial training* (based on *robust optimization).* It means training on simulated adversarial examples.

Preventing adversarial examples

Certified defenses

Ensure that no example can exist inside a ball with radius the norm used for the perturbation

Detect suspicious queries

Identify deviations from expected on distribution of successive queries from a client

⇒ Unclear whether they are really effective.

0.1 Takeaways on adversarial examples

Protecting ML against output alteration (by adversarial examples) is **HARD**

In the adversary controls the inputs in deployed models, they always win!

Unclear whether there will ever be effective defenses. High-dimensional spaces difficult to characterize

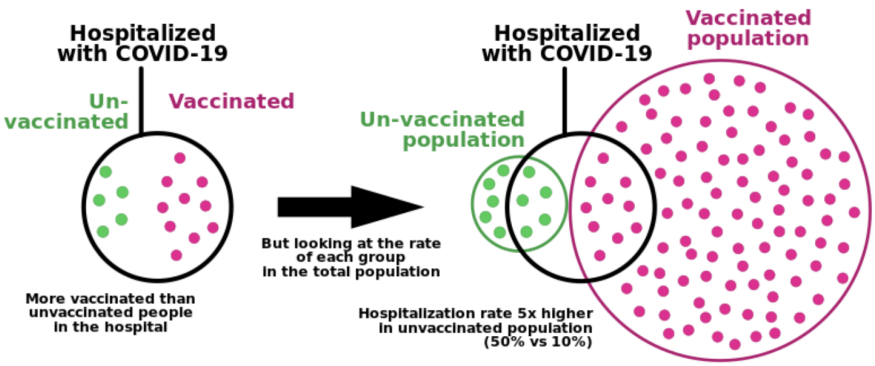
Adversarial training is the most promising protection technique.

> If adversarial examples can occur for an ML usage you are considering, be very careful because this can create a severe vulnerability

Biases and Fallacies

Base Rate Fallacy

We assume equal distribution of cases, if they are not falsely interpret results.



Prosecutor's fallacy

Quick reminder: AI performance metrics

Example: Hate speech detection.

"Only 1 in a million people would match this DNA profile. Therefore, there's only a 1-in-a-million chance the defendant is innocent."

`P(match | innocent)` = 1/1,000,000 ≠ `P(innocent | match) = ?`

Distributional Shift

Distribution of data is not uniform! August 2018 Accuracy on Easy Pilot Parliaments Benchmark in performance on Facial Analysis Task of Gender Classification just works better for lighter males because of data unbalance.

Controversial ML research

When the algorithm was presented with two photos where one picture was definitely of a gay man and the other heterosexual, it was able to determine which was which 81% of the time. With women, the figure was 71%

What did the algorithm learn?

Faces or stereotypical poses/gestures in the Dating site Facebook pictures used for training

Would it work in other social networks? Does it work evenly for different races? And for different social groups?

A Bigger Problem: Bias Reinforcement

Skewed results that overfit because prediction may be biased.

Statistical bias: difference between an estimator expected value and the true value Very limited! Nothing about errors, nothing about distributional shift Text

Group fairness: outcome should not differ between demographic groups Predictive parity: same prediction regardless of group (aka, Calibration) Equal false positive (rates) Equal false negative (rates) ...

Individual fairness: similar? individuals should be treated similarly?